UNITED STATES DEPARTMENT OF THE INTERIOR

GEOLOGICAL SURVEY

A TI-59 Calculator Program for Computation of Schlumberger Resistivity Sounding Curve for Models with as many as 25 Horizontal Layers

by

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Although this program has been extensively tested, the USGS cannot guarantee that it will give correct results in any or all particular applications.

TITLE Schlumberger Resistivity Sounding

PROGRAMMER Ahmed Samir M. El Kahwagy* DATE 10/20/80

Partitioning (Op 17) 2,3,9,8,9

This program was developed to do rapid computation of theoretical Schlumberger resistivity sounding curves. It is based on the application of Ghosh's linear filter (Ghosh, 1971) to resistivity transforms calculated for a given earth model.

Haines and Campbell (1980) wrote a program for the computation of Schlumberger and Wenner sounding curves over structures with as many as 10 horizontal layers. Although it is generally not possible to visually recognize more than 8 layers on a field curve, the present program can be used to compute sounding curves from digitized electric logs provided the number of layers does not exceed 25.

The input data consists of the thicknesses, the resistivities, and the number of layers (up to 25). The program will compute the apparent resistivities at the rate of 6 points per logarithmic cycle starting at abscissa value of AB/2 of 1 and ending at 10,000. The computed apparent resistivities will be stored successively in registers 37 to 61. This represents the output of the program, and the contents of these registers can be recalled by the user.

Note: The dimensions of the computed electrode spacings (AB/2) and apparent resistivities will correspond to those of the entered thicknesses and resistivities, respectively.

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The following is the output of computed apparent resistivities for the above-mentioned illustrating example.

Computed apparent resistivities	Stored in registers	
241.1431472	37	- for AB/2=1
217.0136337	38	
179.1268944	39	
139.846785	40	
113.1147091	41	
100.9726029	42	
96.86244975	43	- for AB/2=10
95.1592856	44	
93.71288423	45	
91.59566606	46	
86.95311787	47	
76.90352711	48	
61.51721231	49	- for AB/2=100
46.82094031	50	
40.00588464	51	
43.1819453	52	
53.71142519	53	
68.20108221	54	
84.45314038	55	- for AB/2=1000
100.9135485	56	
116.0076031	57	
128.363842	58	
137.3269032	59	
143.100719	60	
146.4431868	61	- for AB/2=10000

References

Ghosh, D. P., 1971, Inverse filter coefficients for computation of apparent resistivity standard curves for a horizontally stratified earth, Geophys. Prospect., 19, p. 769-775.

Haines, D. N. and Campbell, D. L., 1980, Texas Instruments Model 59 Hand Calculator Program to calculate theoritical Wenner and Schlumberger vertical electrode soundings of a structure of up to 10 horizontal layers, U.S. Geological Survey Open-File Report 80-190, 15 p.

USER INSTRUCTIONS

STEP	PROCEDURE			ENTER	PRESS			DISPLAY
1	Clear memory				2nd	CMs		
2	Repartition memory storage area			9	2nd	Op	17	239.89
3	Load Side 1 of Card Load Side 2 of Card				INV	2nd	Write	1. 2.
4	Input layer parameters: starting with the							
	first layer, enter and store thicknesses							
	and resistivities as shown in the following							
	example of a 4-layer model:							
		Thickness	Resistivity					
	First Layer	1	259	1.00259	STO	12		1.00259
	Second Layer	46	94	46.00094	STO	13		46.00094
	Third Layer	150	27	150.00027	STO	14		150.00027
	Fourth Layer	---	150	0.00150	STO	15		0.00150
	There are 25 available registers, from no.							
	12 to no. 36, for layer parameters entry.							
	The last input resistivity value represents							
	the resistivity of the infinitely thick							
	last layer.							
5	Enter number of layers (N=4 for above example)						4	4.
6	Press A to start program						A	
7	End of program is indicated by the display of 61							

Press RCL followed by a register number from 37 to 61 to display successively the computed apparent resistivities for AB/2 of 1, 1.46, 2.15, 3.16, 4.64, 6.81, 10, 14.6, ..., 10000.

Operating Restriction

The values of layer resistivities should not be larger than 99999. A six digit number will be considered as a decimal fraction. Layer thicknesses can be entered up to a 4-digit number.

USER DEFINED KEYS	DATA REGISTERS (INV Ins)		LABELS (Op 08)
A Starting of program		80 1.467799268	INV Ins CE CLR x⇄t x²
B		81	√x 1/x STO RCL SUM yˣ
C	12-36 for input	82	EE ( ) ÷ GTO ×
D	layer parameter	83 filter	SBR − RST + R/S •
E		84 coefficients	+/− = CLR INV ln log
A'		85	Exc Prd x⇄t √x Exp 10ˣ
B'	(for recalling	86	Deg Pause x² 1/x yˣ STO
C'	37-61 computed appar-	87	Lbl x≥t Σ+ x̄,s DSZ STO
D'	78 100 rent resistivity	88	sin cos tan Int Abs RST
E'	79 100000	89	x≥0 Pol

FLAGS	- 0	- 1	- 2	- 3	- 4	- 5	- 6	7	8	9

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PROGRAMMER _____ DATE _____

LOC	CODE	KEY	COMMENTS
000	76	LBL	
001	11	A	start, store no. of layers
002	42	STO	
003	01	01	
004	43	RCL	max AB/2
005	79	79	
006	32	X:T	
007	43	RCL	min. AB/2 and store
008	80	80	
009	33	X^2	
010	55	÷	
011	43	RCL	
012	78	78	
013	95	=	
014	42	STO	
015	02	02	
016	03	3	
017	07	7	
018	42	STO	
019	09	09	clear register
020	25	CLR	
021	42	STO	
022	10	10	
023	71	SBR	to recall and stor S and h
024	01	01	
025	05	05	
026	43	RCL	
027	02	02	next AB/2
028	65	×	
029	43	RCL	
030	80	80	
031	95	=	
032	42	STO	
033	02	02	
034	77	GE	Test for max. AB/2
035	00	00	
036	42	42	
037	69	DP	if no, go
038	29	29	to next
039	61	GTO	AB/2
040	00	00	
041	20	20	
042	03	3	
043	07	7	
044	42	STO	
045	03	03	
046	42	STO	
047	04	04	
048	25	CLR	
049	08	8	store first filter coefficient
050	01	1	
051	42	STO	
052	05	05	
053	25	CLR	clear for
054	42	STO	
055	10	10	summation
056	08	8	
057	09	9	
058	32	X:T	
059	43	RCL	
060	04	04	
061	42	STO	
062	03	03	
063	73	RC*	convolve resistivity transform with filter coefficients
064	05	05	
065	65	×	
066	73	RC*	
067	03	03	
068	95	=	
069	44	SUM	
070	10	10	
071	69	DP	
072	23	23	
073	69	DP	
074	23	23	
075	43	RCL	test for conv. if no continue, if yes go 085
076	05	05	
077	67	EQ	
078	00	00	
079	85	85	
080	69	DP	
081	25	25	
082	61	GTO	
083	00	00	
084	63	63	
085	43	RCL	store apparent resistivity
086	10	10	
087	72	ST*	
088	04	04	
089	06	6	limit of computation
090	01	1	
091	32	X:T	
092	43	RCL	test for limit, if yes stop.
093	04	04	
094	67	EQ	
095	01	01	
096	03	03	If no, go back and Continue
097	69	DP	
098	24	24	
099	61	GTO	
100	00	00	
101	48	48	
102	68	NOP	
103	91	R/S	
104	00	0	
105	01	1	
106	01	1	
107	85	+	
108	43	RCL	
109	01	01	
110	95	=	
111	42	STO	
112	08	08	
113	73	RC*	
114	08	08	
115	22	INV	
116	59	INT	
117	65	×	
118	43	RCL	
119	79	79	
120	95	=	
121	42	STO	
122	10	10	
123	69	DP	
124	38	38	
125	43	RCL	
126	01	01	
127	42	STO	
128	00	00	
129	97	DSZ	
130	00	00	
131	01	01	
132	35	35	
133	69	DP	
134	38	38	
135	73	RC*	
136	08	08	
137	59	INT	
138	42	STO	
139	11	11	
140	73	RC*	
141	08	08	
142	22	INV	
143	59	INT	
144	65	×	
145	43	RCL	
146	79	79	
147	95	=	
148	42	STO	
149	07	07	
150	71	SBR	to compute transforms
151	01	01	
152	62	62	
153	97	DSZ	decrement no. of layers
154	00	00	
155	01	01	If zero store transform, If no, Continue
156	33	33	
157	43	RCL	
158	10	10	
159	72	ST*	

store P_n, recall and store resistivity and thickness of $(n-1)$ layer seperately.

Counter for no. of layers

MERGED CODES

62 [INV] [SBR]	72 [STO] [ST*]	83 [GTO] [NOP]
63 [INV] [SBR]	73 [RCL] [RC*]	84 [LBL] [SUM]
64 [INV] [SUM]	74 [SUM] [ST*]	92 [INV] [SBR]

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PROGRAMMER DATE

TI Programmable Coding Form

LOC	CODE	KEY	COMMENTS
160	09	09	
161	92.	RTN	
162	02	2	
163	65	×	
164	43	RCL	
165	11	11	
166	65	×	
167	43	RCL	
168	02	02	
169	35	1/X	
170	95	=	
171	94	+/-	
172	22	INV	
173	23	LNX	
174	42	STO	
175	06	06	
176	53	(	
177	53	(	
178	01	1	
179	75	-	
180	43	RCL	
181	06	06	
182	54	)	
183	55	÷	
184	53	(	
185	01	1	
186	85	+	
187	43	RCL	
188	06	06	
189	54	)	
190	54	)	
191	42	STO	
192	06	06	
193	65	×	
194	43	RCL	
195	07	07	
196	95	=	
197	42	STO	
198	06	06	
199	85	+	
200	43	RCL	
201	10	10	
202	95	=	
203	42	STO	
204	05	05	
205	53	(	
206	01	1	
207	85	+	
208	43	RCL	
209	06	06	
210	65	×	
211	43	RCL	
212	10	10	
213	55	÷	
214	43	RCL	

Compute resistivity transforms from $T_i = \frac{\rho_1 + \rho_1 v_i}{1 + \rho_1 v_i \rho_1 / \rho_i^2}$, $v_i = \frac{1 - e^{-2\pi i u}}{1 + e^{-2\pi i u}}$

LOC	CODE	KEY	COMMENTS
215	07	07	
216	33	X²	
217	54	)	
218	22	INV	
219	49	PRD	
220	05	05	
221	48	EXC	
222	05	05	
223	42	STO	
224	10	10	
225	92	RTN	

Store in Registers

	100.	78
	100000.	79
	1.467799268	80
Filter coefficients	0.0148	81
	-0.0814	82
	0.4018	83
	-1.5716	84
	1.972	85
	0.1854	86
	0.1064	87
	-0.0499	88
	0.0225	89

MERGED CODES

62 [A'] [INT] 72 [STO] [ADV] 83 [GTO] [IND]
63 [SIN] [COS] 73 [RCL] [SUM] 84 [LN] [EXC]
64 [TAN] [PRD] 74 [SUM] [INT] 92 [INV] [SBR]

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